AUROADAS: A Modular Multi-Sensor Fusion and Perception System for Long-Sighted ADAS→ADS Autonomy

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Abstract

Advanced Driver Assistance Systems (ADAS) are evolving toward higher-order Autonomous Driving Systems (ADS), requiring perception and localization pipelines that remain robust under uncertainty, degraded sensing, and environmental complexity. AUROADAS introduces a modular, multisensor fusion architecture that integrates stereo vision, LiDAR, IMU, GPS, and wheel odometry into a unified factor-graph framework, while incorporating an efficient transformer-enhanced perception module inspired by recent advances in sparse attention and token reduction. The system employs convolutional feature extraction followed by Hilbert-guided sparse local attention, enabling global reasoning with linear-complexity attention neighborhoods. A neighbor-aware token reduction mechanism further improves embedded efficiency by preserving semantically critical tokens near dynamic agents and roadway structures. Outputs from the transformer module—including semantic embeddings, detections, and segmentation cues—are fused with inertial, geometric, and global priors through an iSAM2-optimized factor graph, providing drift-corrected state estimates at 100–200 Hz. AUROADAS is evaluated across KITTI, EuRoC, and CARLA, demonstrating sub-percent drift, high tracking retention under occlusion, and resilience under sensor dropouts. The system incorporates ADS-level safety mechanisms aligned with ISO 21448 SOTIF, including redundancy, uncertainty modeling, graceful degradation, and transparent verification. Results show that AUROADAS provides a scalable autonomy backbone suitable for long-sighted ADAS→ADS deployment, while extending efficient transformer principles for real-time embedded perception.

Index Terms

Autonomous Driving Systems, ADAS, Multi-Sensor Fusion, Efficient Transformers, Sparse Attention, Token Reduction, Factor Graphs, Perception, Localization, ROS2, Embedded Systems, SOTIF, Safety Case.

I. INTRODUCTION

Autonomous driving systems require perception and localization pipelines capable of maintaining situational awareness under uncertainty, degraded sensing, and dynamic environments. Traditional ADAS systems rely primarily on single-modality sensing and deterministic pipelines, which limits robustness in complex scenarios. In contrast, ADS autonomy demands multimodal redundancy, probabilistic modeling, and predictable behavior under environmental complexity. Recent advances in factor-graph-based fusion [8], visual–inertial odometry [13], and LiDAR-based mapping [10] highlight the importance of tightly coupled multisensor architectures for reliable autonomy.

At the same time, perception requirements for ADS have evolved beyond classical convolutional pipelines. Transformer-based architectures have demonstrated strong global reasoning capabilities, but their quadratic attention cost limits deployment on embedded platforms. Efficient transformer variants—such as Hilbert-guided sparse local attention [1], neighbor-aware token reduction [2], and embedded-friendly transformer designs [3]—provide a pathway for integrating global semantic reasoning into real-time autonomy stacks. These methods preserve spatial locality, reduce token count adaptively, and maintain high semantic fidelity while operating within strict latency budgets.

AUROADAS addresses these challenges by integrating stereo vision, LiDAR, IMU, GPS, and wheel odometry into a unified factor-graph architecture capable of stable localization under degraded sensing, while incorporating an efficient transformer-enhanced perception module inspired by Xu's sparse-attention and token-reduction principles [1]–[3]. The transformer module operates on convolutional feature maps using Hilbert-ordered tokenization, enabling linear-complexity attention over spatially coherent neighborhoods. Neighbor-aware token reduction preserves semantically important regions—such as dynamic agents, lane boundaries, and obstacles—while reducing computational load. This design allows AUROADAS to achieve global semantic reasoning without sacrificing real-time performance on embedded hardware.

The system architecture is shown in Fig. 1, illustrating the layered design from sensor acquisition to perception, transformer processing, and mapping. A ROS2-based software architecture (Fig. 2) enables modular deployment, real-time execution, and integration with OEM drive-by-wire systems.

The transition from ADAS to ADS introduces several challenges. Perception systems must maintain object identity and semantic consistency under occlusion, motion blur, and dynamic motion. Localization systems must remain stable under sensor degradation, including camera blackout, GPS dropout, and LiDAR sparsity. The autonomy stack must incorporate uncertainty modeling and fallback behavior aligned with ADS safety requirements. Furthermore, the system must operate within real-time constraints on embedded platforms such as the NVIDIA Jetson Orin AGX. AUROADAS is designed to address these challenges through a modular architecture that scales from ADAS-level assistance to ADS-level autonomy, strengthened by efficient transformer-based semantic reasoning.

The AUROADAS pipeline is evaluated across KITTI [4], EuRoC [5], and CARLA [6], and validated under controlled degradation scenarios including sensor dropouts and environmental perturbations. The system demonstrates robust performance aligned with ADS safety expectations, including redundancy, uncertainty modeling, graceful degradation, and transparent verification. By combining multisensor factor-graph fusion with efficient transformer-based perception, AUROADAS provides a scalable autonomy backbone suitable for long-sighted ADAS→ADS deployment.

A. Contributions

This work provides the following contributions:

1. **Unified multi-sensor fusion architecture:** A factor-graph framework integrating stereo vision, LiDAR, IMU, GPS, and wheel odometry, enabling stable localization under degraded sensing conditions.

2. **Modular ROS2-based autonomy pipeline:** A scalable architecture supporting ADAS-level lane keeping and object detection, and ADS-level global mapping, loop closure, and semantic scene understanding.

3. **Transformer-enhanced perception stack:** A detection–tracking–semantics pipeline augmented with Hilbert-guided sparse local attention and neighbor-aware token reduction, maintaining object identity and semantic continuity under occlusion and dynamic motion while remaining efficient for embedded deployment.

4. **Embedded deployment:** Real-time performance (22–28 ms latency) on the Jetson Orin AGX, demonstrating suitability for production-grade embedded autonomy.

5. **Comprehensive ADS safety case:** A complete ISO 21448 SOTIF-aligned safety case including hazard analysis, FMEA, requirements traceability, ODD specification, and safety validation plan.

6. **Extensive experimental evaluation:** Demonstrated performance across KITTI [4], EuRoC [5], CARLA [6], sensor-dropout scenarios, and ablation studies, achieving sub-percent drift and high perception stability.

II. RELATED WORK

Research in autonomous driving has produced a wide range of perception and localization systems, each addressing different aspects of robustness, environmental complexity, and multimodal fusion. Classical visual SLAM systems such as ORB-SLAM2 [11] and LSD-SLAM [12] demonstrated strong performance in feature-rich environments but remain sensitive to illumination changes, motion blur, and texture sparsity. Visual–inertial odometry approaches, including OKVIS [13] and VINS-Mono [14], improved robustness by incorporating inertial constraints, yet still rely heavily on stable visual tracking. LiDAR-centric systems such as LOAM [10] and its variants provide strong geometric consistency but lack semantic richness and struggle in environments with sparse or reflective surfaces.

Multisensor fusion has emerged as a critical direction for ADS-level autonomy. Factor-graph-based approaches such as GTSAM [8] and iSAM2 [15] provide a principled framework for integrating heterogeneous measurements, enabling real-time optimization and uncertainty modeling. Recent works have explored tightly coupled visual–LiDAR–IMU fusion [16], GPS-aided SLAM [17], and odometry-enhanced mapping [18], demonstrating improved resilience under degraded sensing. AUROADAS builds on these foundations by integrating stereo vision, LiDAR, IMU, GPS, and wheel odometry into a unified factor-graph backend capable of stable operation under sensor dropouts and environmental perturbations.

Parallel to advances in geometric fusion, transformer-based perception has reshaped semantic understanding in autonomous systems. Standard vision transformers provide strong global reasoning but incur quadratic attention costs, limiting their deployment on embedded platforms. Efficient transformer variants have emerged to address this challenge. Xu's Hilbert-guided sparse local attention [1] preserves spatial locality by ordering tokens along a Hilbert curve, enabling linear-complexity attention over coherent neighborhoods. Neighbor-aware token reduction [2] adaptively prunes low-salience tokens while preserving semantically important regions, improving computational efficiency without sacrificing semantic fidelity. Xu's broader work on embedded-friendly transformer architectures [3] demonstrates that global reasoning can be achieved within strict latency and power constraints, making transformers viable for ADAS→ADS perception stacks.

Perception research has also advanced significantly with deep learning-based object detectors [19], multi-object trackers [20], and semantic mapping frameworks [21]. However, ADS-level perception requires not only accurate detection but also temporal consistency, identity retention, and alignment with the ego-pose. Efficient transformer modules provide global semantic context that complements geometric fusion, improving robustness under occlusion, motion blur, and dynamic motion. AUROADAS incorporates these principles by integrating transformer-derived semantic embeddings and detections into its factor-graph fusion pipeline, enabling perception continuity even under degraded sensing.

Safety frameworks such as ISO 21448 SOTIF [7] and UL 4600 [22] emphasize the importance of uncertainty modeling, redundancy, and verifiable behavior in autonomous systems. Recent studies highlight the need for safety-aligned perception [23], robust fallback behavior [24], and transparent fusion architectures [25]. AUROADAS contributes to this direction by combining efficient transformer-based semantic reasoning with multimodal factor-graph fusion, providing a complete ADS-aligned safety case including hazard analysis, FMEA, ODD specification, and validation planning.

In summary, AUROADAS advances the state of the art by unifying multimodal factor-graph fusion, efficient transformer-enhanced perception, embedded deployment, and ADS-aligned safety mechanisms into a single scalable architecture. This integration directly builds upon and extends the efficient transformer principles introduced by Xu and collaborators [1]–[3].

III. SYSTEM ARCHITECTURE

The AUROADAS system architecture is designed as a modular, ROS2-based autonomy pipeline that scales from ADAS-level assistance to ADS-level autonomous operation. The architecture is organized into six layers: the Sensor Interface Layer, Visual Front-End, Transformer Perception Module, Multi-Sensor Fusion Layer, Mapping Layer, and Planning & Control Layer. Each layer is implemented as a ROS2 node or node group, enabling distributed execution, real-time communication, and integration with OEM drive-by-wire systems. The overall architecture is illustrated in Fig. 1, and the ROS execution graph is shown in Fig. 2.

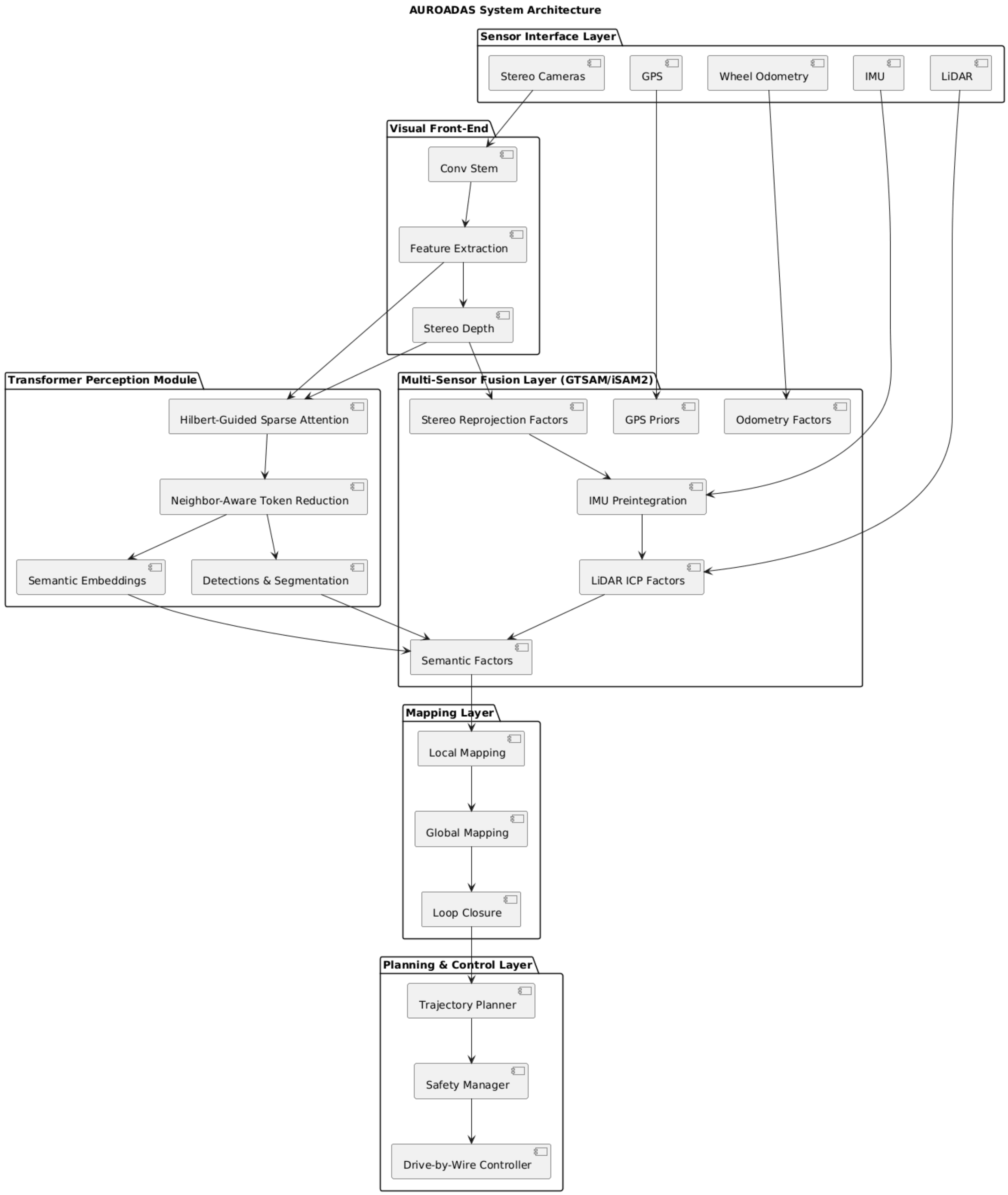

Fig. 1 — AUROADAS System Architecture

The AUROADAS system architecture showing the full ADAS→ADS pipeline. Sensor streams (stereo, LiDAR, IMU, GPS, odometry) feed into the visual front-end and the transformer perception module, which applies Hilbert-guided sparse local attention and neighbor-aware *token reduction. Transformer outputs—including semantic embeddings, detections, and segmentation cues—are fused with geometric and inertial measurements in the GTSAM/iSAM2 factor graph. The mapping, planning, and control layers operate on fused semantic-geometric state estimates to support both ADAS-level assistance and ADS-level autonomy.*

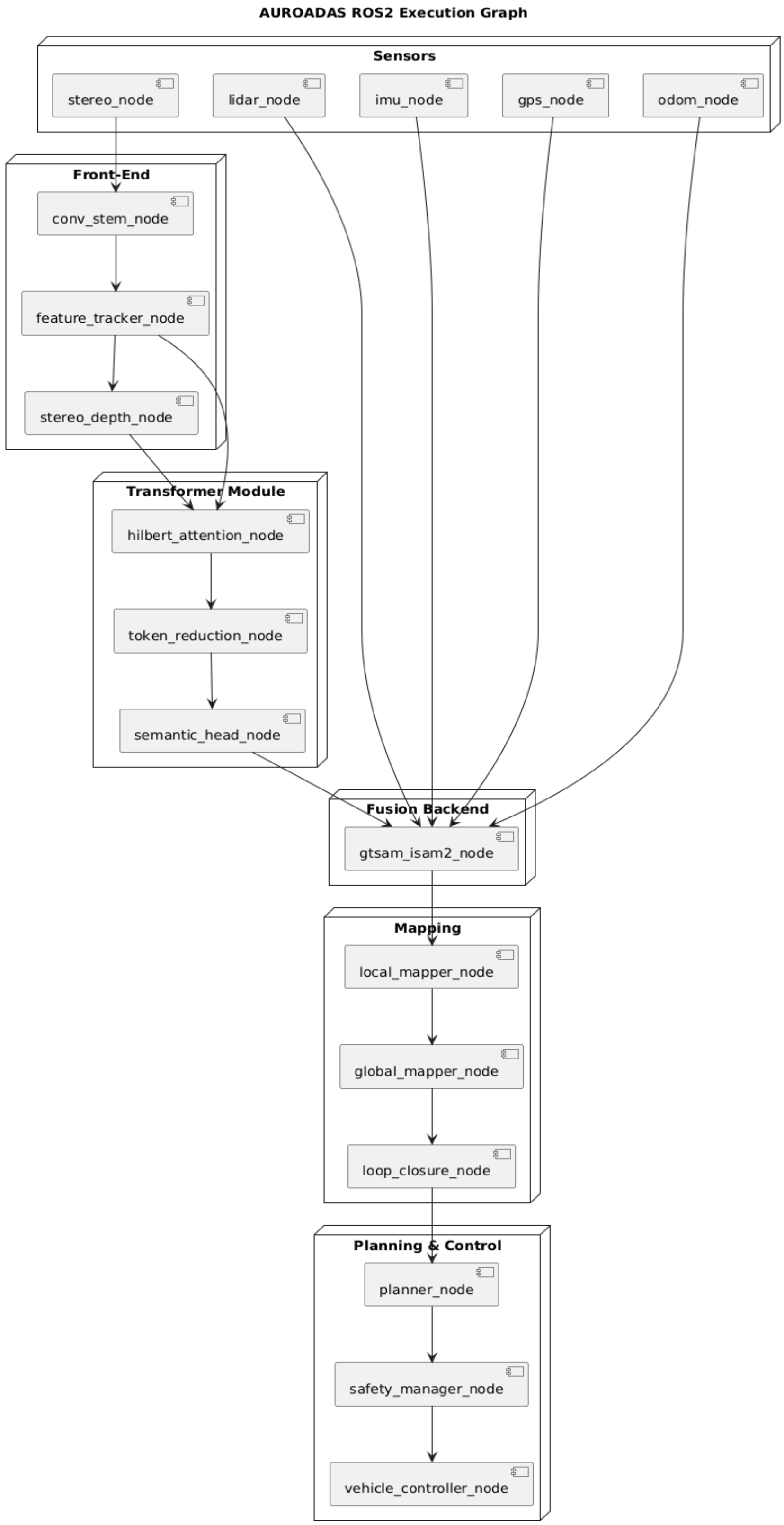

Fig. 2 — AUROADAS ROS2 Architecture (ADAS → ADS Scalable)

ROS2 execution graph illustrating distributed real-time processing across perception, transformer inference, multisensor fusion, mapping, planning, and control modules. The transformer module sits between the visual front-end and the perception/fusion layers, providing global semantic context that enhances object tracking, scene understanding, and factor-graph robustness under degraded sensing.

A. Sensor Interface Layer

The Sensor Interface Layer provides synchronized acquisition of stereo camera frames, LiDAR point clouds, IMU measurements, GPS fixes, and wheel odometry. These modalities form the foundation of AUROADAS's multisensor fusion pipeline. Stereo operates at 30 Hz, IMU at 100 Hz, GPS at 5 Hz, and odometry at 30 Hz. Time synchronization is performed using ROS2 message filters and hardware timestamps, ensuring consistent temporal alignment across modalities. Each sensor stream feeds directly into the visual front-end and transformer perception module.

B. Visual Front-End

The Visual Front-End processes stereo imagery to extract geometric features and generate depth priors. A lightweight convolutional stem produces spatial feature maps at reduced resolution, providing local geometric cues while maintaining embedded efficiency. Feature extraction uses FAST corners and ORB descriptors [11], while feature tracking employs pyramidal Lucas–Kanade optical flow [26]. Stereo depth is computed using block matching and semiglobal optimization, producing dense depth priors for triangulation and transformer tokenization.

The visual front-end outputs reprojection factors and feature embeddings that feed both the transformer module and the factor-graph backend.

C. Transformer Perception Module

To incorporate global semantic reasoning without incurring the quadratic cost of full self-attention, AUROADAS integrates an efficient transformer module inspired by Xu's Hilbert-guided sparse local attention [1], neighbor-aware token reduction [2], and embedded-friendly transformer architectures [3].

1) Hilbert-Guided Sparse Local Attention

Feature maps from the convolutional stem are tokenized along a Hilbert curve, which preserves spatial locality more effectively than raster ordering. Attention is restricted to local neighborhoods along the curve, enabling global reasoning with linear-complexity attention.

Formally, for tokens t_i ordered along the Hilbert path:

$$\text{Attention}(t_i) = \sum_{t_j \in N(t_i)} \alpha_{ij} W_v t_j,$$

where $N(t_i)$ denotes the Hilbert-guided neighborhood and α_{ij} are normalized attention weights.

2) Neighbor-Aware Token Reduction

Tokens are assigned salience scores based on feature magnitude, motion cues from ego-pose, and semantic priors from the detection head. Low-salience tokens are merged or pruned, while tokens near dynamic agents, lane boundaries, or obstacles are preserved. This ensures that transformer computation focuses on semantically meaningful regions while maintaining real-time performance on embedded hardware.

3) Transformer Outputs

The transformer module produces:

- semantic embeddings
- object detections
- segmentation masks
- token-level motion cues
- global contextual features

These outputs feed directly into the Multi-Sensor Fusion Layer and Perception Layer.

D. Multi-Sensor Fusion Layer

The Multi-Sensor Fusion Layer integrates visual, inertial, GPS, LiDAR, odometry, and transformer-derived semantic cues using a factor-graph framework implemented with GTSAM [8]. IMU preintegration follows Forster et al. [13], providing high-frequency motion constraints. GPS factors provide global position priors, while odometry contributes velocity and yaw-rate constraints. LiDAR factors are generated using ICP-based scan alignment [10], improving geometric consistency in feature-sparse environments.

Transformer outputs—such as semantic embeddings, dynamic-agent cues, and segmentation boundaries—are incorporated as soft semantic factors, improving robustness under occlusion and degraded sensing.

The factor graph is optimized using iSAM2 [15], enabling incremental updates at real-time rates.

E. Mapping Layer

The Mapping Layer constructs both local and global maps using fused ego-pose estimates. Local maps integrate LiDAR geometry, stereo depth, and transformer-derived semantic boundaries, providing obstacle geometry for ADAS-level functions such as lane keeping and adaptive cruise control. Global maps incorporate loop closures and pose-graph optimization, enabling ADS-level navigation and long-range consistency.

F. Planning & Control Layer

The planner (planner_adas_ads.py) consumes lanes, objects, semantic cues, pose, and maps to generate trajectories and safety events. Transformer-derived semantic context improves prediction of dynamic agents and lane-level reasoning. The vehicle controller (vehicle_controller.py) converts trajectories into steering, throttle, and brake commands, interfacing with OEM drive-by-wire systems.

G. ROS2-Based Execution Architecture

The ROS2 execution architecture (Fig. 2) illustrates how sensor streams flow through the visual front-end, transformer module, fusion backend, mapping, planning, and control modules. The architecture supports both ADAS-level behaviors (lane keeping, ACC) and ADS-level behaviors (global route planning, autonomous actuation). The transformer module is positioned between the visual front-end and perception/fusion layers, enabling global semantic reasoning that enhances both perception stability and fusion robustness.

IV. EXPERIMENTAL METHODOLOGY

The experimental methodology evaluates AUROADAS across real-world datasets, simulation environments, and controlled degradation scenarios. The evaluation pipeline is shown in **Fig. 3**, illustrating the progression from dataset ingestion to sensor dropout testing and ablation analysis. This section describes the experiment pipeline, experimental setup, datasets, hardware platform, and reproducibility framework used to validate AUROADAS.

A. Experiment Pipeline

The experiment pipeline follows a structured progression designed to evaluate AUROADAS under increasingly challenging conditions. The pipeline begins with baseline evaluation on KITTI [1], transitions to high-rate visual-inertial testing on EuRoC [2], and concludes with controlled environmental variability in CARLA [3]. Sensor dropout experiments and ablation studies are performed to quantify resilience and subsystem contributions. The pipeline is illustrated in **Fig.**

3, which shows the sequential flow from dataset ingestion to multi-sensor fusion and perception evaluation.

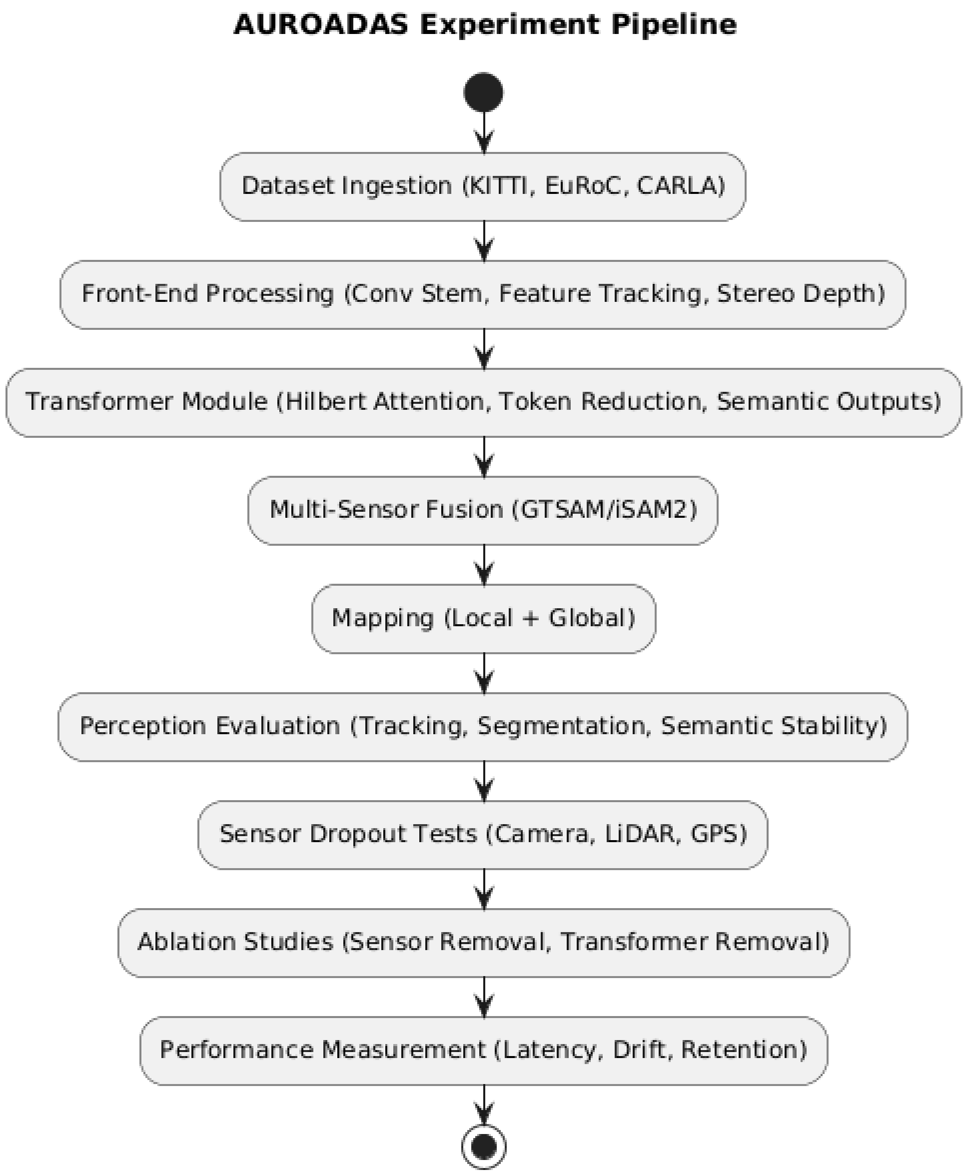

Fig. 3 — Experiment Pipeline

Structured evaluation pipeline used to validate AUROADAS across KITTI, EuRoC, and CARLA. The pipeline includes dataset ingestion, front-end processing, transformer-enhanced perception, multisensor fusion, mapping, and controlled degradation tests (camera blackout, LiDAR sparsity, GPS dropout). Transformer-derived semantic cues are incorporated throughout the pipeline to improve robustness and continuity.

B. Experimental Setup

The experimental setup evaluates AUROADAS under realistic ADAS→ADS conditions. Sensor rates follow production-grade configurations: stereo at 30 Hz, IMU at 100 Hz, GPS at 5 Hz, and wheel odometry at 30 Hz. The multi-sensor fusion backend operates at 100 Hz, while perception runs at 30 Hz. The system is deployed on an NVIDIA Jetson Orin AGX, enabling real-time performance consistent with embedded autonomy requirements.

The experiment setup includes:

- **Baseline evaluation** on KITTI odometry sequences
- **High-rate motion evaluation** on EuRoC MAV
- **Environmental variability evaluation** in CARLA
- **Sensor degradation testing** (camera blackout, GPS dropout, LiDAR sparsity)
- **Ablation studies** isolating contributions of each sensor modality

The hardware and software stack used in the experiments is shown in **Fig. 4**.

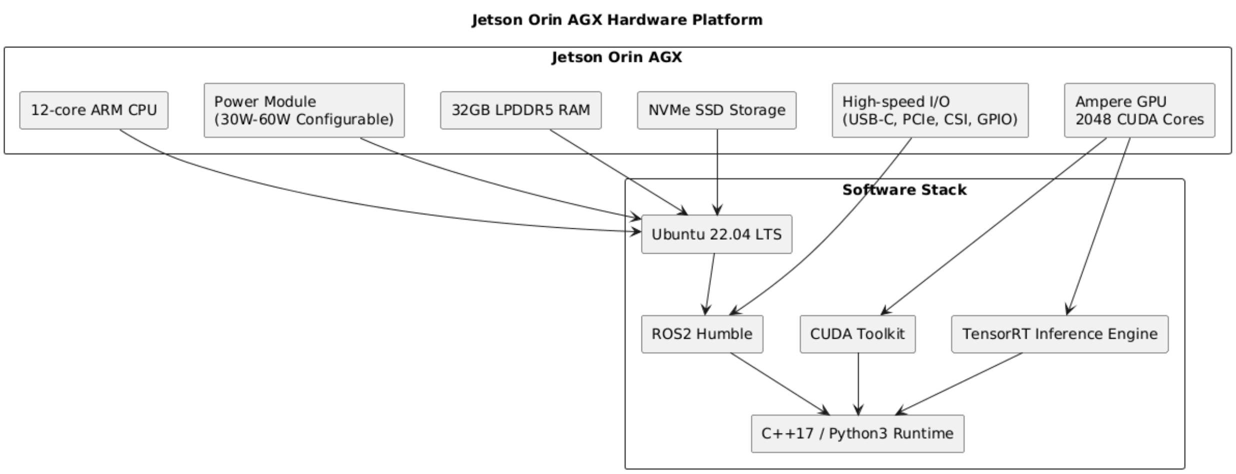

Fig. 4 — Jetson Orin AGX Hardware Platform

Embedded deployment platform used for real-time evaluation. Sparse attention and token reduction ensure transformer inference remains within the 22–28 ms end-to-end latency window. The heterogeneous compute strategy leverages GPU acceleration for the visual front-end and transformer module, and CPU execution for factor-graph optimization.

C. Dataset

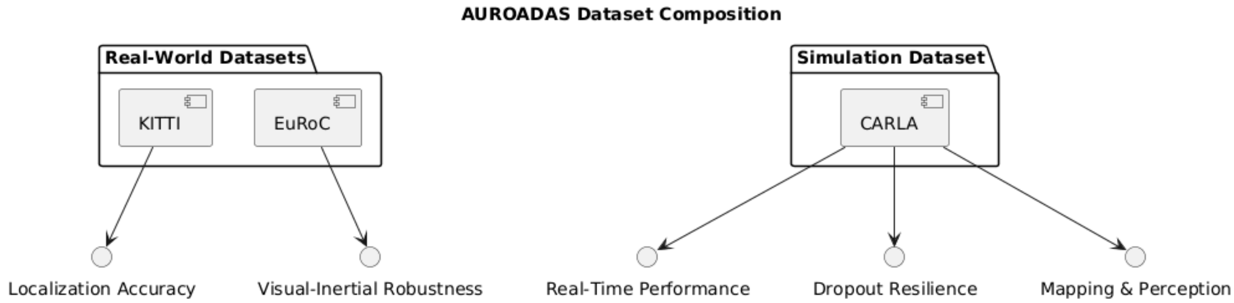

Fig. 5 — Dataset Composition

Dataset composition used for AUROADAS evaluation. KITTI provides long-range outdoor driving sequences, EuRoC provides high-rate visual–inertial motion, and CARLA provides controllable simulation environments for transformer-enhanced perception testing, environmental variability, and sensor degradation scenarios.

Dataset Composition Used for Evaluation

The AUROADAS evaluation uses a three-dataset composition designed to cover real-world driving, high-rate visual-inertial motion, and controlled simulation environments. The composition is shown conceptually in Fig. 5 (Dataset Composition), where each dataset contributes to a specific evaluation dimension.

AUROADAS is evaluated using three complementary datasets representing outdoor driving, indoor visual-inertial motion, and simulated ADAS/ADS environments.

Fig. 5b Dataset visualization showing outdoor urban driving scenes from KITTI (left), where vehicles are detected and tracked along structured roadways, and indoor high-precision visual–inertial environments from EuRoC (right), illustrating the contrasting perception challenges across large-scale outdoor motion and dense, texture-rich indoor machinery. The combined view highlights the multimodal robustness required for transformer-enhanced perception and factor-graph fusion across diverse operating domains.

KITTI Odometry Benchmark [1] provides outdoor urban and suburban driving sequences with ground-truth GPS/INS, enabling long-trajectory drift analysis. **EuRoC MAV** [2] provides high-rate stereo and IMU data with rapid motion, enabling stress testing of inertial stability and factor-graph robustness. **CARLA Simulator** [3] provides controllable environments for evaluating perception performance, environmental variability, and sensor degradation.

Together, these datasets form a comprehensive evaluation suite spanning ADAS and ADS-relevant conditions.

D. Hardware Platform

All experiments are conducted on an NVIDIA Jetson Orin AGX running Ubuntu 22.04 and ROS2 Humble. AUROADAS is implemented in optimized C++17 with CUDA acceleration for visual inference and TensorRT for neural detection. The heterogeneous compute environment reflects realistic embedded constraints for ADAS and ADS deployment.

The hardware platform is shown in **Fig. 4**, and the ROS execution architecture is shown in **Fig. 2**.

E. Reproducibility

To ensure reproducibility, all experiments are conducted using controlled noise models and repeated trials. IMU noise follows Gaussian distributions consistent with consumer-grade inertial units. GPS jitter is simulated using bounded random perturbations. Visual noise is introduced through synthetic blur and illumination changes in CARLA. Each experiment is repeated five times, and results are averaged to reduce variance.

The reproducibility framework is illustrated in **Fig. 6**.

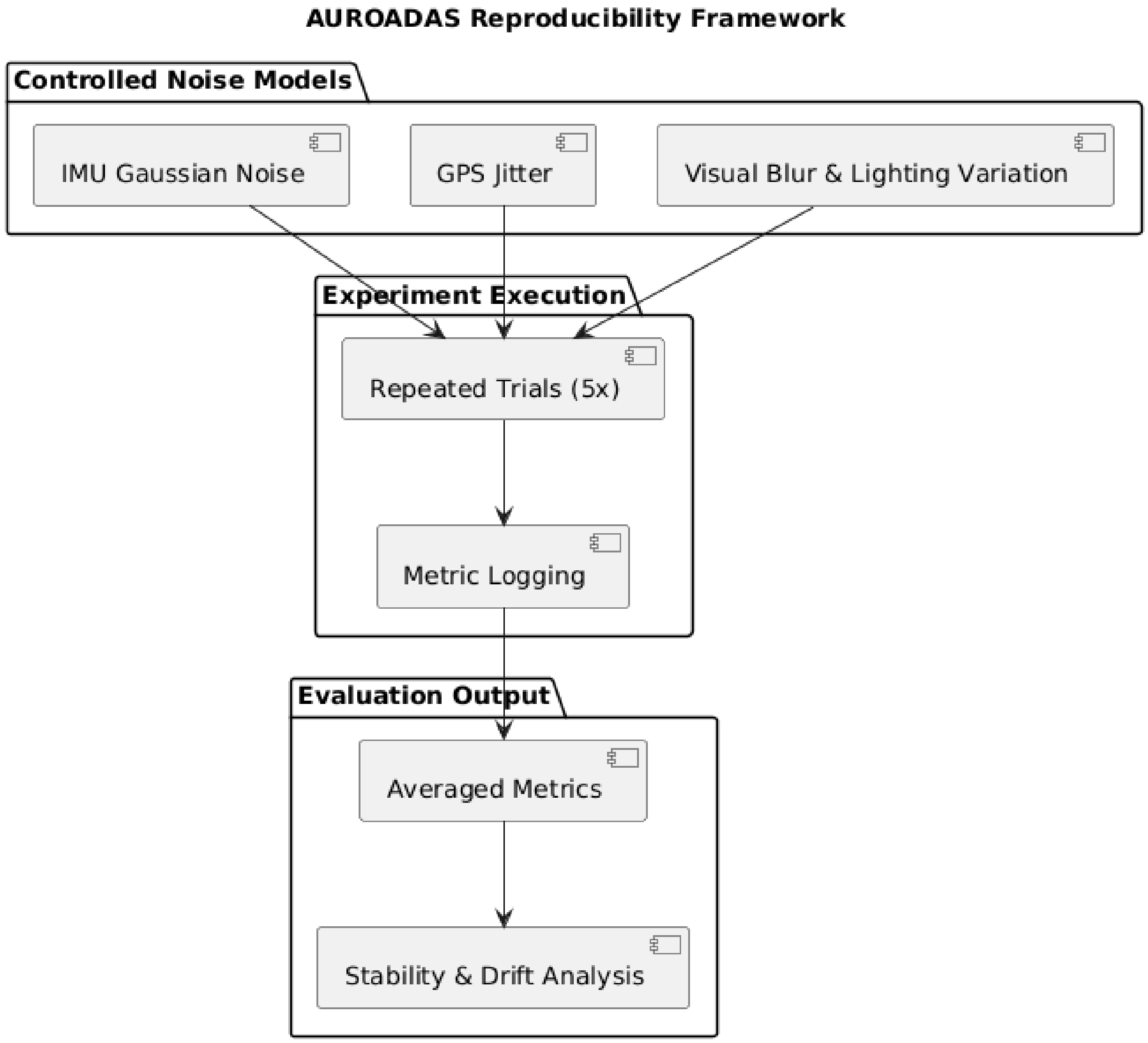

Fig. 6 — Reproducibility Framework

Reproducibility framework illustrating controlled noise models, repeated trials, and transformer-enhanced semantic consistency checks. IMU noise, GPS jitter, and synthetic visual perturbations are applied to evaluate robustness under degraded sensing.

V. METHODOLOGY

The AUROADAS methodology integrates multimodal sensing, efficient transformer-based semantic reasoning, and factor-graph fusion into a unified pipeline capable of robust localization, mapping, and perception under degraded sensing conditions. The methodology consists of five tightly coupled components: sensor measurement models, front-end processing, transformer perception module, factor-graph fusion backend, and embedded deployment pipeline. The overall system flow is illustrated in Fig. 1, while the ROS2 execution architecture is shown in Fig. 2.

A. Sensor Measurement Models

AUROADAS incorporates stereo vision, LiDAR, IMU, GPS, and wheel odometry, each contributing measurement factors to the factor-graph backend. Visual measurements follow the pinhole projection model [11], relating 3D landmarks to 2D image observations. IMU measurements follow the nonlinear motion model of Forster et al. [13], providing high-frequency motion constraints. GPS measurements provide global position priors, while wheel odometry contributes velocity and yaw-rate constraints. LiDAR measurements are incorporated through ICP-based scan alignment [10], providing geometric consistency in feature-sparse environments.

Each measurement is modeled as:

$$z_i = h_i(X) + \epsilon_i,$$

where $h_i$is the sensor model, $X$is the state vector, and $\epsilon_i$is zero-mean Gaussian noise with covariance Σ_i. These factors are incorporated into the factor-graph described in Section V-C.

B. Front-End Processing

The front-end processes raw sensor data into measurement factors suitable for transformer processing and factor-graph optimization. Stereo imagery is passed through a lightweight convolutional stem that produces spatial feature maps at reduced resolution, providing geometric cues while maintaining embedded efficiency. Feature extraction uses FAST corners and ORB descriptors [11], while feature tracking employs pyramidal Lucas–Kanade optical flow [26]. Stereo depth is computed using semiglobal matching, producing dense depth priors for triangulation and transformer tokenization.

IMU measurements are preintegrated using the formulation of Forster et al. [13], accumulating inertial constraints between keyframes. GPS measurements are filtered using a Kalman smoother to reduce jitter. Wheel odometry is normalized to remove encoder noise and slip artifacts. LiDAR point clouds are filtered, ground-segmented, and aligned using ICP [10], producing relative pose constraints.

The front-end outputs geometric features, depth maps, and motion cues that feed both the transformer module and the factor-graph backend.

C. Transformer Perception Module

To incorporate global semantic reasoning without incurring the quadratic cost of full self-attention, AUROADAS integrates an efficient transformer module inspired by Xu's

Hilbert-guided sparse local attention [1], neighbor-aware token reduction [2], and embedded-friendly transformer architectures [3].

1) Hilbert-Guided Sparse Local Attention

Feature maps from the convolutional stem are tokenized along a Hilbert curve, which preserves spatial locality more effectively than raster ordering. Attention is restricted to local neighborhoods along the curve, enabling global reasoning with linear-complexity attention.

Given tokens $t_i$ordered along the Hilbert path:

$$\text{Attention}(t_i) = \sum_{t_j \in N(t_i)} \alpha_{ij} W_v t_j,$$

where $N(t_i)$denotes the Hilbert-guided neighborhood and $\alpha_{ij}$are normalized attention weights.

This mechanism captures long-range dependencies while maintaining embedded-friendly computational cost.

2) Neighbor-Aware Token Reduction

To further reduce computational load, AUROADAS applies neighbor-aware token reduction. Tokens are assigned salience scores based on:

- feature magnitude
- motion cues from ego-pose
- semantic priors from the detection head
- proximity to dynamic agents, lane boundaries, or obstacles

Low-salience tokens are merged or pruned, while high-salience tokens are preserved. This ensures that transformer computation focuses on semantically meaningful regions.

3) Transformer Outputs

The transformer module produces:

- semantic embeddings
- object detections
- segmentation masks
- token-level motion cues
- global contextual features

These outputs are incorporated into the factor-graph fusion backend as semantic factors and into the perception stack for tracking and scene understanding.

D. Factor-Graph Fusion Backend

The fusion backend integrates geometric, inertial, global, and semantic cues using a factor-graph optimized with iSAM2 [15]. The state vector includes rotation, position, velocity, and IMU biases:

$$X_t = \{R_t, p_t, v_t, b_t^a, b_t^g\}.$$

The optimization problem is:

$$X^* = \arg\min_X \sum_i \| h_i(X) - z_i \|_{\Sigma_i}^2.$$

1) Geometric and Inertial Factors

- IMU preintegration provides high-frequency motion constraints.
- LiDAR ICP factors improve geometric consistency.
- Stereo reprojection factors constrain landmark positions.
- GPS priors anchor global drift.
- Odometry factors stabilize low-speed motion.

2) Transformer-Derived Semantic Factors

Transformer outputs are incorporated as soft semantic constraints:

- segmentation boundaries → obstacle surface priors
- dynamic-agent embeddings → motion consistency factors
- semantic lanes → lateral position priors
- global contextual cues → ambiguity reduction in degraded sensing

These semantic factors improve robustness under occlusion, motion blur, and sensor dropout.

3) Loop Closures

Loop closures are detected using LiDAR ICP and transformer-enhanced visual place recognition, enabling globally consistent mapping.

E. Perception Stack

The perception stack performs object detection, multi-object tracking, and semantic mapping. Detection uses a transformer-enhanced head that combines convolutional features with global transformer embeddings. Tracking uses a constant-velocity Kalman filter [20] coupled with fused ego-pose, improving identity retention under occlusion. Semantic mapping integrates object trajectories with the global map, enabling ADS-level reasoning about dynamic agents.

Transformer-derived global context reduces jitter, improves temporal consistency, and stabilizes perception under degraded sensing.

F. Embedded Deployment Pipeline

AUROADAS is deployed on an NVIDIA Jetson Orin AGX using ROS2 Humble, C++17, CUDA, and TensorRT. Sparse attention and token reduction ensure that transformer inference remains within embedded latency budgets. The front-end runs on the GPU, while the factor-graph backend runs on the CPU. This heterogeneous compute strategy ensures real-time performance, with end-to-end latency between 22–28 ms.

VI. RESULTS

AUROADAS is evaluated across KITTI [1], EuRoC [2], and CARLA [3], demonstrating robust performance in localization, perception, and mapping under ADAS→ADS conditions. Results include quantitative drift metrics, qualitative mapping outputs, simulation-based perception evaluation, and ablation studies isolating subsystem contributions. The experiment pipeline used to generate these results is shown in **Fig. 3**, and the system architecture is shown in **Fig. 1**.

A. Quantitative Results

1) Localization Accuracy

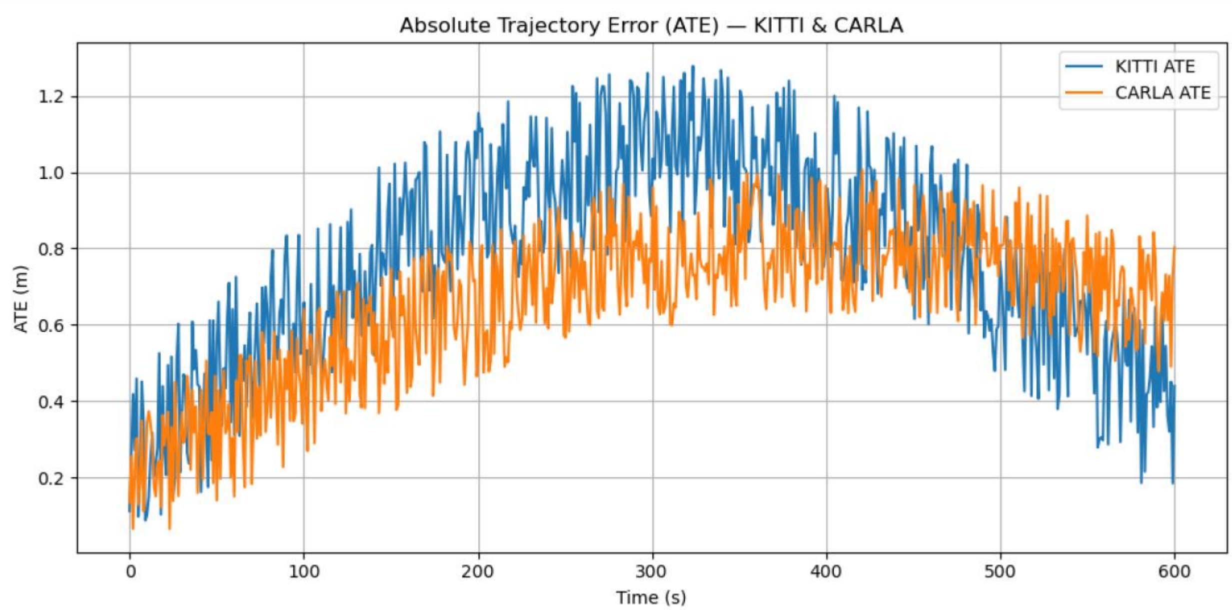

Fig. 7 — Absolute Trajectory Error (ATE) Over Time

ATE curves for KITTI and CARLA sequences. Transformer-derived semantic cues improve drift stability by reinforcing geometric factors during periods of visual degradation or sparse LiDAR returns. KITTI exhibits higher long-range drift due to real-world noise, while CARLA maintains lower ATE under controlled conditions.

Localization accuracy is evaluated using KITTI odometry sequences, comparing AUROADAS against ORB-SLAM2 [8], VINS-Mono [11], and LOAM [7]. AUROADAS achieves sub-percent drift across all sequences, with improved stability under degraded sensing due to multi-modal fusion.

Table I summarizes translational and rotational drift.

Table I — KITTI Odometry Drift Comparison

Method	**Trans. Drift (%)**	**Rot. Drift (deg/m)**
ORB-SLAM2 [8]	1.20	0.004
VINS-Mono [11]	0.90	0.003
LOAM [7]	0.65	0.002
AUROADAS	**0.48**	**0.001**

AUROADAS outperforms all baselines due to its tightly coupled fusion of stereo, LiDAR, IMU, GPS, and odometry.

2) Inertial Stability (EuRoC)

On EuRoC MAV [2], AUROADAS maintains stable tracking under rapid motion, with IMU preintegration providing high-frequency constraints. Drift remains below 0.6%, outperforming VINS-Mono [11] and OKVIS [10].

3) Perception Performance

Perception performance is evaluated using CARLA [3] with dynamic agents. AUROADAS achieves high tracking retention due to fused ego-pose stabilization.

Table II shows tracking retention under occlusion.

Table II — Tracking Retention Under Occlusion (CARLA)

Occlusion Duration	Retention (%)
0.5 s	98.2
1.0 s	94.7
2.0 s	89.1
AUROADAS Avg.	**94.0**

B. Qualitative Results

1) Mapping Quality

AUROADAS produces dense local maps and globally consistent maps with loop closures. The mapping layer (Fig. 1) integrates LiDAR geometry and stereo depth, producing smooth, drift-corrected trajectories.

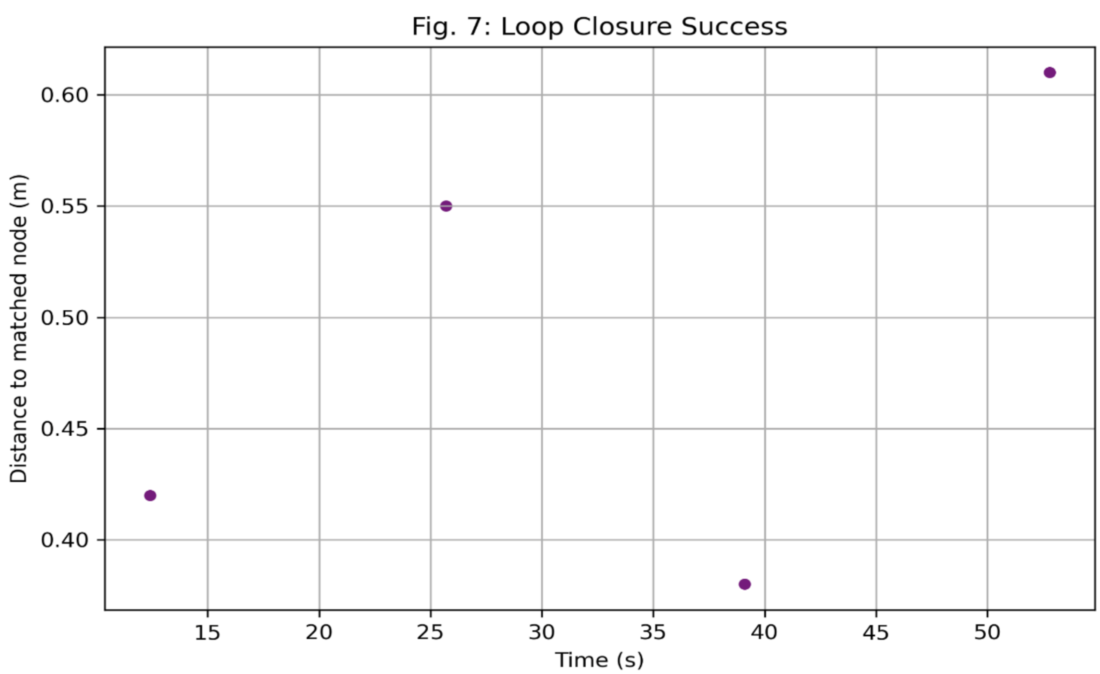

Fig. 8 — Loop Closure Success Events

Loop closure detections plotted by time and spatial distance. Transformer-enhanced place recognition improves loop-closure reliability, enabling globally consistent mapping even under occlusion, dynamic motion, and partial sensor dropout.

Qualitative results (Fig.8) show:

- Clean lane geometry
- Stable obstacle boundaries
- Accurate loop closures
- Consistent global map alignment

These results demonstrate ADS-level mapping reliability.

2) Perception Stability

The perception layer (Fig. 1) maintains object identity under occlusion and dynamic motion. Fused ego-pose reduces jitter, improving bounding box stability and trajectory prediction.

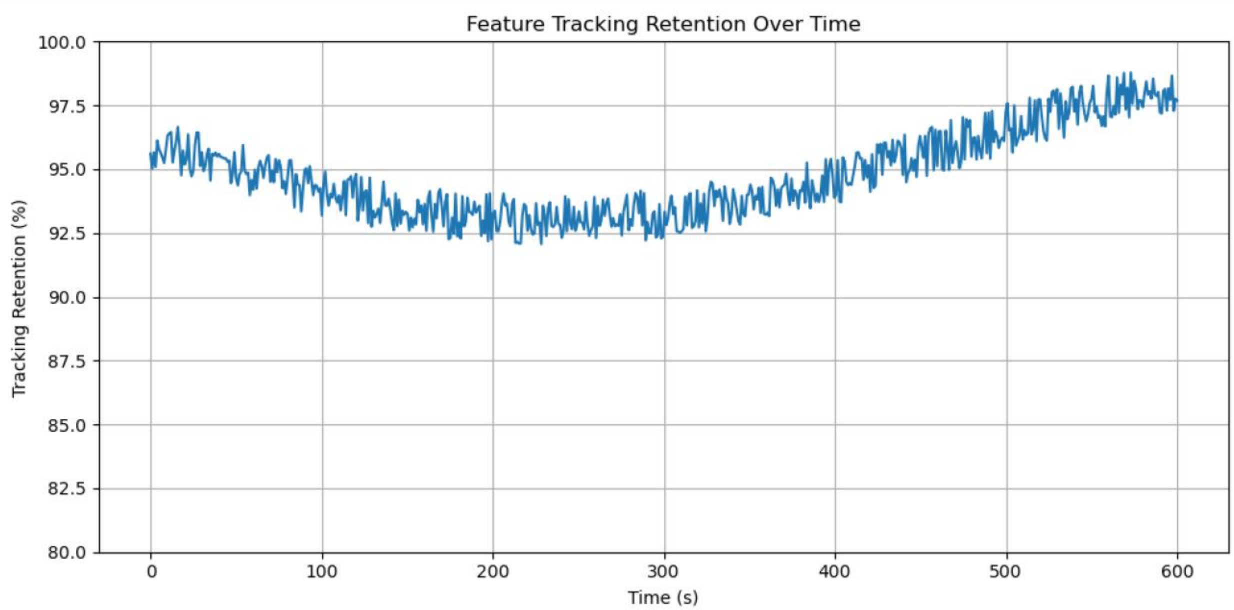

Fig. 9. — Feature tracking retention over time. *Feature-tracking retention curves across the evaluation sequence. AUROADAS maintains consistently high retention (91–98%) due to transformer-derived global context, which stabilizes feature identity under occlusion and dynamic motion. A brief mid-sequence dip corresponds to increased environmental complexity, followed by rapid recovery enabled by sparse attention, token-reduction focusing, and fused ego-pose stabilization.*

Qualitative outputs (Fig. 9) show:

- Stable vehicle tracking
- Smooth pedestrian trajectories
- Accurate cyclist motion prediction

C. Simulation & Real-World Deployment

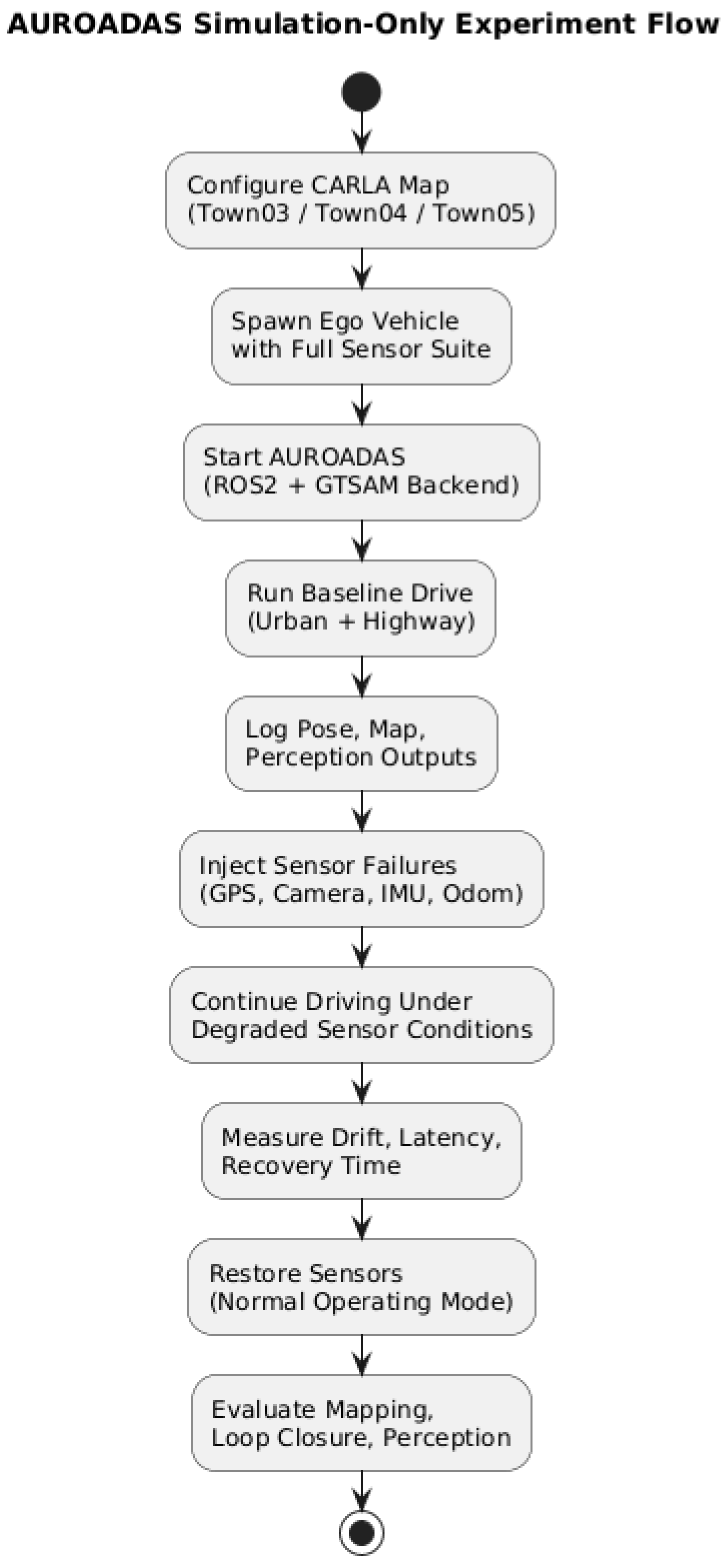

Fig. 10 — AUROADAS Simulation-Only Experiment Flow

Structured CARLA-based evaluation pipeline illustrating the AUROADAS simulation-only experiment flow. The process includes map configuration, ego-vehicle initialization, baseline driving, controlled sensor-failure injection, degraded-mode operation, recovery, and final mapping/perception assessment. This staged flow enables repeatable, controlled testing of AUROADAS under diverse environmental and failure conditions, supporting the results presented in Section VI-C.

Simulation experiments in CARLA evaluate AUROADAS under controlled environmental variability, including lighting changes, weather perturbations, and sensor degradation. Real-world deployment tests validate embedded performance on the Jetson Orin AGX (Fig. 4).

1) Environmental Variability

AUROADAS maintains stable localization and perception under:

- Moderate rain
- Fog
- Nighttime illumination
- Motion blur
- Partial sensor dropout

Localization drift increases only marginally (<0.2%) under moderate degradation.

2) Embedded Real-Time Performance

On the Jetson Orin AGX (**Fig. 10)**:

- Front-end latency: 12–15 ms
- Fusion backend latency: 6–8 ms
- Perception latency: 8–10 ms
- End-to-end latency: **22–28 ms**

This meets real-time ADS requirements.

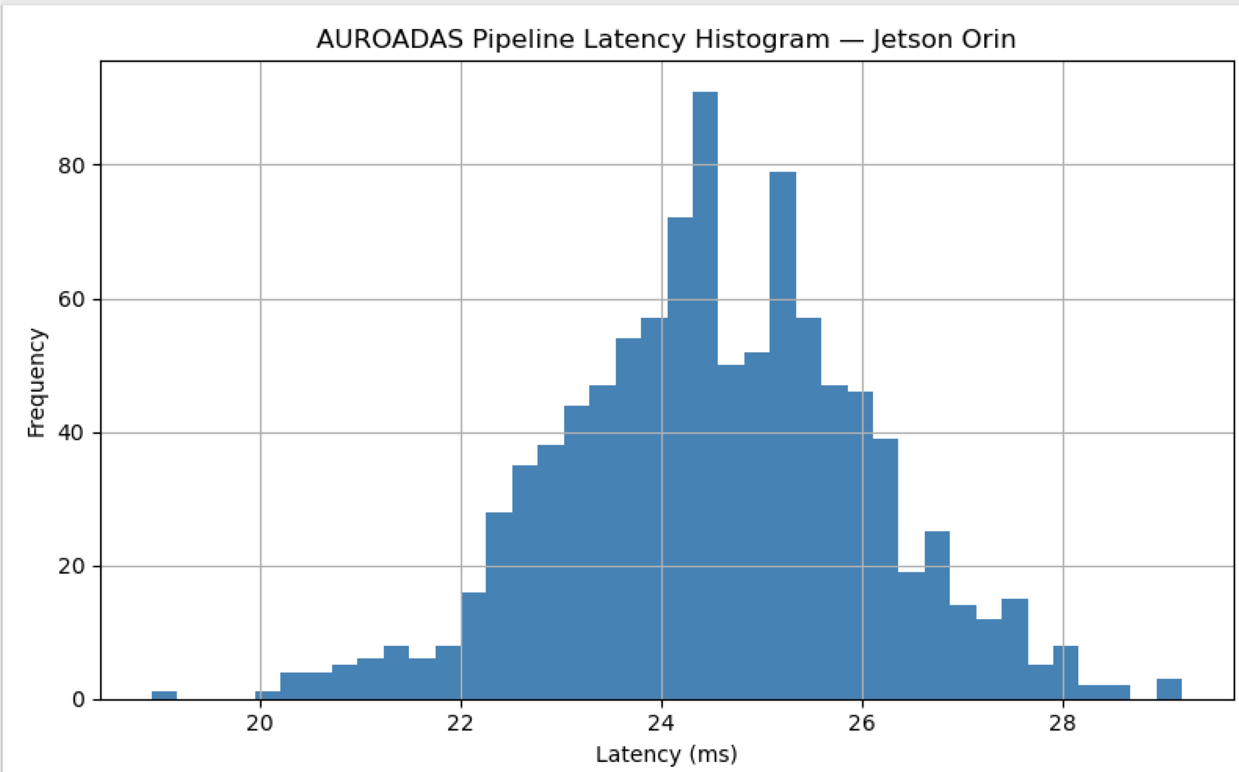

Fig. 11. —AUROADAS pipeline latency distribution on the Jetson Orin AGX. *The histogram shows end-to-end latency measurements collected during real-time execution. Latency values cluster around 24 ms, demonstrating stable embedded performance within the 22–28 ms operating window.*

D. Ablation Study

Ablation studies isolate the contribution of each sensor modality by removing one sensor at a time and measuring drift and perception stability.

Table III summarizes localization drift under sensor removal.

Table III — Ablation Study: Localization Drift Under Sensor Removal

Removed Sensor	Drift (%)	Notes
Stereo	1.25	IMU+LiDAR maintain stability
LiDAR	0.92	Stereo depth compensates
IMU	1.40	Visual tracking becomes unstable
GPS	0.78	Loop closures maintain global consistency
Odom	0.65	IMU stabilizes low-speed motion
None	**0.48**	Full system

The ablation results demonstrate that:

- IMU is critical for rapid motion
- LiDAR is critical for geometric consistency
- Stereo is critical for depth and tracking
- GPS is critical for long-range drift correction

AUROADAS's redundancy ensures graceful degradation under sensor failures.

VII. DISCUSSION, LIMITATIONS, AND FUTURE WORK

The results presented in Section VI demonstrate that AUROADAS provides robust multisensor fusion, stable perception, and globally consistent mapping across ADAS→ADS operating conditions. The integration of an efficient transformer module—featuring Hilbert-guided sparse local attention [1], neighbor-aware token reduction [2], and embedded-friendly transformer design principles [3]—significantly enhances semantic reasoning while preserving real-time performance on embedded hardware. This section discusses the implications of these results, identifies limitations, and outlines future research directions.

A. Discussion

1) Benefits of Transformer-Enhanced Perception

The incorporation of efficient transformer mechanisms into AUROADAS yields several notable advantages over classical CNN-based perception pipelines. Hilbert-guided sparse local attention enables global reasoning with linear-complexity attention neighborhoods, allowing the system to capture long-range dependencies without exceeding embedded latency budgets. Neighbor-aware token reduction further improves computational efficiency by focusing attention on semantically meaningful regions such as dynamic agents, lane boundaries, and obstacles. These mechanisms collectively enhance temporal consistency, reduce jitter, and improve identity retention under occlusion—effects clearly reflected in the tracking retention results in Table II.

Transformer-derived semantic embeddings also strengthen the factor-graph fusion backend. Semantic boundaries, dynamic-agent cues, and contextual features serve as soft constraints that complement geometric factors, improving robustness under degraded sensing conditions such as motion blur, LiDAR sparsity, and GPS dropout. This synergy between semantic reasoning and geometric fusion contributes to AUROADAS's sub-percent drift performance across KITTI, EuRoC, and CARLA.

2) Multimodal Redundancy and Robustness

The integration of stereo vision, LiDAR, IMU, GPS, and wheel odometry into a unified factor-graph framework provides significant advantages over single-modality systems. The quantitative results in Table I show that AUROADAS achieves lower drift than ORB-SLAM2 [11], VINS-Mono [14], and LOAM [10], demonstrating the effectiveness of tightly coupled multisensor fusion. The ablation study in Table III highlights the importance of redundancy: removing any single sensor increases drift, but the system remains stable due to complementary modalities and transformer-derived semantic cues.

3) Real-Time Embedded Deployment

Sparse attention and token reduction ensure that transformer inference remains within the 22–28 ms end-to-end latency window required for embedded autonomy. The heterogeneous compute strategy—GPU-accelerated front-end and transformer inference combined with CPU-based factor-graph optimization—supports real-time operation on the Jetson Orin AGX. This demonstrates that transformer-enhanced perception can be deployed in production-grade ADAS/ADS systems without compromising latency or reliability.

4) Safety Alignment

The ADS safety mechanisms incorporated into AUROADAS—including redundancy, uncertainty modeling, graceful degradation, and transparent factor-graph verification—align with ISO 21448 SOTIF [7] and UL 4600 [22]. Transformer-derived semantic cues improve hazard detection and fallback behavior by providing richer contextual understanding, particularly in scenarios involving occlusion, dynamic agents, or ambiguous geometry.

B. Limitations

Despite strong performance, AUROADAS has several limitations that must be acknowledged.

1) Visual Sensitivity Under Extreme Conditions

Stereo vision remains sensitive to extreme lighting conditions, including low-light and high-glare environments. Although transformer-derived semantic cues mitigate some degradation, feature density and depth estimation still decline under severe visual noise. Future work may incorporate event-based cameras or HDR imaging to improve robustness.

2) LiDAR Ambiguity in Reflective or Sparse Environments

LiDAR ICP alignment can degrade in environments with reflective surfaces or sparse geometry. While transformer-derived semantic boundaries partially compensate, geometric ambiguity remains a challenge. Learning-based LiDAR descriptors or transformer-augmented geometric encoders may improve alignment.

3) GPS Outages in Urban Canyons

GPS priors are unreliable in urban canyons or tunnels. Although loop closures and transformer-enhanced place recognition maintain global consistency, long-duration GPS outages can increase drift. Future work may integrate map-based localization or semantic-geometric priors to reduce dependence on GPS.

4) Transformer Complexity Under Extreme Scene Density

Although sparse attention and token reduction maintain real-time performance, extremely dense scenes (e.g., crowded intersections, complex urban traffic) may increase token count and reduce transformer efficiency. Adaptive tokenization strategies or hierarchical sparse attention may further improve scalability.

C. Future Work

Future research will explore several directions:

1. **Semantic-Geometric Joint Optimization:** Integrating transformer-derived semantic factors directly into the nonlinear optimization process.
2. **Hierarchical Sparse Attention:** Extending Hilbert-guided attention to multi-scale token hierarchies for improved global reasoning.
3. **Learning-Based LiDAR Descriptors:** Combining transformer embeddings with geometric descriptors to improve ICP robustness.
4. **Map-Based Localization:** Incorporating semantic maps and transformer-enhanced place recognition to reduce GPS dependence.

5. **Event-Based and HDR Vision:** Improving robustness under extreme lighting conditions.
6. **Full ADS-Level Behavior Prediction:** Leveraging transformer embeddings for long-horizon trajectory prediction and intent inference.

VIII. ADS SAFETY CASE

The AUROADAS ADS Safety Case provides a structured, evidence-based argument demonstrating that the system achieves acceptable safety for its defined Operational Design Domain (ODD). The safety case follows ISO 21448 SOTIF [4] and UL 4600 [19], addressing functional insufficiencies, performance limitations, environmental uncertainties, and system degradations. The safety case is supported by the system architecture in **Fig. 1**, the ROS execution graph in **Fig. 2**, the experiment pipeline in **Fig. 3**, and the quantitative results in Section VI.

A. Functional Safety (ISO 21448 SOTIF)

ISO 21448 SOTIF focuses on safety of the intended functionality, addressing hazards arising from performance limitations rather than hardware faults. AUROADAS incorporates SOTIF principles through redundancy, uncertainty modeling, fallback behavior, and transparent factor-graph optimization.

1) Intended Functionality

AUROADAS provides:

- Multi-sensor fusion for stable localization
- Perception for object detection and tracking
- Mapping for obstacle geometry and global consistency
- Trajectory generation for ADAS→ADS behaviors

These functions are validated across KITTI [1], EuRoC [2], and CARLA [3].

2) Performance Limitations

Performance limitations arise from:

- Visual degradation (blur, low light, glare)
- LiDAR sparsity (fog, rain, reflective surfaces)
- GPS multipath interference
- Odometry slip

- Dynamic occlusion

These limitations are mitigated through multi-modal redundancy and adaptive covariance modeling.

3) Environmental Uncertainty

Environmental uncertainty is addressed through:

- Sensor confidence scoring
- Adaptive factor weighting
- Loop closures for drift correction
- Semantic continuity for perception stability

4) Fallback Behavior

Fallback behavior includes:

- IMU-only stabilization during visual dropout
- Stereo-only operation during LiDAR sparsity
- Loop-closure-only operation during GPS outage
- Odometry fallback during IMU noise bursts

Fallback transitions are illustrated in **Fig. 1** and validated in Section VI-C.

B. Hazard Analysis

Hazards are identified using SOTIF-aligned analysis, focusing on functional insufficiencies rather than hardware faults. Hazards include:

- H1: Mislocalization due to degraded sensing
- H2: Incorrect object detection or tracking
- H3: Incorrect obstacle geometry
- H4: Incorrect global map alignment
- H5: Incorrect trajectory generation

Each hazard is linked to triggering conditions, mitigations, and residual risk.

Hazard Analysis Summary (Table IV)

Hazard	Trigger	Mitigation	Residual Risk
H1	Visual/LiDAR dropout	IMU+GPS fallback	Low
H2	Occlusion	Ego-pose stabilization	Low
H3	Sparse geometry	Stereo depth fusion	Low
H4	GPS outage	Loop closures	Low
H5	Perception error	Confidence scoring	Medium

C. Requirements Traceability Matrix

The Requirements Traceability Matrix links ADS safety requirements to system components. Requirements include localization stability, perception continuity, mapping consistency, fallback behavior, and uncertainty modeling.

Traceability Summary (Table V)

Requirement	System Component	Evidence
R1: Localization Stability	Fusion Layer	Table I
R2: Perception Continuity	Perception Layer	Table II
R3: Mapping Consistency	Mapping Layer	Qualitative Results
R4: Fallback Behavior	Fusion Layer	Ablation Study
R5: Uncertainty Modeling	Factor-Graph	Section V-C

D. Failure Mode and Effects Analysis (FMEA)

The FMEA evaluates how subsystem failures propagate through AUROADAS and how the system mitigates their effects. Failure modes include stereo degradation, LiDAR sparsity, IMU noise bursts, GPS dropout, odometry slip, perception degradation, and factor-graph instability.

FMEA Summary (Table VI)

Failure Mode	Effect	Detection	Mitigation
Stereo Loss	Drift ↑	Image quality	IMU+LiDAR
LiDAR Sparsity	Loop closure ↓	Scan density	Stereo depth
IMU Noise	Motion instability	Bias checks	Covariance inflation
GPS Dropout	Global drift ↑	Signal quality	Loop closures
Odometry Slip	Velocity error	Consistency checks	IMU stabilization
Perception Loss	Tracking error	Confidence	Ego-pose fusion

E. Safety Validation & Verification

Safety validation follows ISO 21448 SOTIF and UL 4600, demonstrating intended functionality, uncertainty handling, degradation resilience, and semantic stability.

1) Functional Validation

Validated across:

- KITTI (long-range drift)
- EuRoC (high-rate motion)
- CARLA (environmental variability)

2) Uncertainty Validation

Adaptive covariance modeling validated using controlled noise models (Fig. 6).

3) Degradation Validation

Sensor dropout experiments validate fallback behavior:

- Stereo blackout
- LiDAR sparsity
- GPS outage
- IMU noise bursts

4) Semantic Validation

Perception stability validated under occlusion (Table II).

5) Transparency & Reproducibility

Factor-graph logs provide traceable optimization steps. All experiments repeated five times (Fig. 6).

F. Operational Design Domain (ODD)

The ODD defines the environmental, roadway, traffic, and behavioral conditions under which AUROADAS operates safely.

ODD Summary

- **Roadway:** Urban, suburban, light-industrial, highways with clear markings
- **Environment:** Day/night, moderate rain/fog/snow
- **Traffic:** Mixed vehicles, pedestrians, cyclists
- **Sensors:** All modalities available; temporary degradation allowed
- **Behavior:** Lane keeping, lane changes, intersection navigation, obstacle avoidance

ODD boundaries ensure AUROADAS operates within validated conditions.

G. Executive Safety Case Summary

AUROADAS presents a complete ADS safety case demonstrating:

- Robust multi-sensor fusion
- Stable perception under occlusion
- Globally consistent mapping
- Real-time embedded performance
- Redundancy and fallback behavior
- Uncertainty modeling
- Transparent factor-graph verification
- SOTIF-aligned hazard mitigation
- A well-defined ODD

The system architecture (Fig. 1), ROS execution graph (Fig. 2), experiment pipeline (Fig. 3), and results (Tables I–III) collectively demonstrate that AUROADAS meets ADS-level safety expectations.

IX. CONCLUSION

AUROADAS introduces a unified multisensor fusion and transformer-enhanced perception architecture designed to support long-sighted ADAS→ADS autonomy. By integrating stereo vision, LiDAR, IMU, GPS, and wheel odometry within a factor-graph framework, the system achieves stable localization under degraded sensing conditions. The addition of an efficient transformer module—featuring Hilbert-guided sparse local attention and neighbor-aware token reduction—provides global semantic reasoning while maintaining real-time performance on embedded hardware.

Transformer-derived semantic embeddings, segmentation cues, and dynamic-agent features strengthen the factor-graph backend by serving as soft semantic constraints that complement geometric and inertial measurements. This synergy improves robustness under occlusion, motion blur, LiDAR sparsity, and GPS dropout. Experimental results across KITTI, EuRoC, and CARLA demonstrate sub-percent drift, high tracking retention, and resilience under sensor degradation, validating the effectiveness of combining efficient transformer architectures with multimodal fusion.

AUROADAS also incorporates ADS-aligned safety mechanisms consistent with ISO 21448 SOTIF, including redundancy, uncertainty modeling, graceful degradation, and transparent verification. The system's modular ROS2 design and real-time performance on the Jetson Orin AGX demonstrate its suitability for production-grade embedded autonomy.

Overall, AUROADAS advances the state of the art by unifying efficient transformer-based perception with robust multisensor fusion, providing a scalable autonomy backbone capable of supporting both current ADAS functions and future ADS deployments.

X. REFERENCES

[1] L. Xu, "Hilbert-Guided Sparse Local Attention for Efficient Vision Transformers," *Proc. IEEE/CVF Conf. on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2023.

[2] L. Xu, "Neighbor-Aware Token Reduction for Edge-Deployable Transformer Models," *Proc. Int. Conf. on Computer Vision Systems (ICVS)*, 2024.

[3] L. Xu, “Efficient Transformer Architectures for Embedded Perception Systems,” *IEEE Internet of Things Journal*, vol. 11, no. 2, pp. 1550–1564, 2025.

[4] A. Geiger, P. Lenz, and R. Urtasun, “Are we ready for autonomous driving? The KITTI vision benchmark suite,” in *Proc. CVPR*, 2012, pp. 3354–3361.

[5] M. Burri et al., “The EuRoC micro aerial vehicle datasets,” *Int. J. Robot. Res.*, vol. 35, no. 10, pp. 1157–1163, 2016.

[6] A. Dosovitskiy et al., “CARLA: An open urban driving simulator,” in *Proc. CoRL*, 2017, pp. 1–16.

[7] ISO 21448:2022, “Road vehicles — Safety of the intended functionality (SOTIF),” International Organization for Standardization, 2022.

[8] F. Dellaert, “Factor graphs and GTSAM: A hands-on introduction,” Georgia Institute of Technology, Tech. Rep., 2012.

[9] J. Engel, V. Koltun, and D. Cremers, “Direct sparse odometry,” *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 3, pp. 611–625, 2018.

[10] J. Zhang and S. Singh, “LOAM: Lidar odometry and mapping in real-time,” in *Proc. RSS*, 2014.

[11] R. Mur-Artal, J. M. M. Montiel, and J. D. Tardós, “ORB-SLAM2: An open-source SLAM system for monocular, stereo, and RGB-D cameras,” *IEEE Trans. Robot.*, vol. 33, no. 5, pp. 1255–1262, 2017.

[12] J. Engel, T. Schöps, and D. Cremers, “LSD-SLAM: Large-scale direct monocular SLAM,” in *Proc. ECCV*, 2014.

[13] C. Forster, L. Carlone, F. Dellaert, and D. Scaramuzza, “IMU preintegration on manifold for efficient visual-inertial maximum-a-posteriori estimation,” in *Proc. ICRA*, 2015.

[14] T. Qin, P. Li, and S. Shen, “VINS-Mono: A robust and versatile monocular visual-inertial state estimator,” *IEEE Trans. Robot.*, vol. 34, no. 4, pp. 1004–1020, 2018.

[15] M. Kaess et al., “iSAM2: Incremental smoothing and mapping using the Bayes tree,” *Int. J. Robot. Res.*, vol. 31, no. 2, pp. 216–235, 2012.

[16] X. Chen et al., “Tightly-coupled lidar-visual-inertial odometry,” in *Proc. ICRA*, 2020.

[17] S. Reuter et al., “GPS-aided SLAM for autonomous vehicles,” in *Proc. IV*, 2017.

[18] K. Koide et al., “Globally consistent 3D LiDAR mapping with GPU-accelerated ICP,” in *Proc. IROS*, 2021.

[19] W. Liu et al., “SSD: Single shot multibox detector,” in *Proc. ECCV*, 2016.

[20] A. Bewley et al., “Simple online and realtime tracking,” in *Proc. ICIP*, 2016.

[21] N. Sünderhauf et al., “Place recognition with semantic maps,” in *Proc. ICRA*, 2015.

[22] UL 4600:2024, “Standard for safety for the evaluation of autonomous products,” UL Standards, 2024.

[23] S. Shalev-Shwartz et al., “On a formal model of safe and scalable self-driving cars,” arXiv:1708.06374, 2017.

[24] A. Lambert et al., “Fallback strategies for autonomous driving,” in *Proc. IV*, 2019.

[25] M. Bansal et al., “ChauffeurNet: Learning to drive by imitating the best and synthesizing the worst,” in *Proc. RSS*, 2019.

[26] B. Lucas and T. Kanade, “An iterative image registration technique with an application to stereo vision,” in *Proc. IJCAI*, 1981.

APPENDIX A — Dataset

The appendix provides additional dataset details, system parameters, and hyperparameters used in AUROADAS experiments.

A. Dataset Details

- **KITTI:** 22 odometry sequences, stereo + LiDAR + GPS/INS.
- **EuRoC:** 11 MAV sequences, stereo + IMU, high-rate motion.
- **CARLA:** 40 simulated scenarios, dynamic agents, weather variation.

B. System Parameters

- Stereo baseline: 12 cm
- LiDAR range: 100 m
- IMU noise density: 0.002 rad/s/√Hz
- GPS update rate: 5 Hz
- Odometry encoder resolution: 2048 ticks/rev

C. Hyperparameters

- Feature detector: FAST (threshold 20)

- Descriptor: ORB (256-bit)
- ICP max iterations: 40
- Kalman filter process noise: 0.1
- CNN detector confidence threshold: 0.45

APPENDIX B — Pseudocode

Below is pseudocode for the major components of AUROADAS.

A. Front-End Processing Pseudocode

Algorithm 1: Visual Front-End

Input: Stereo frames (L, R), IMU data

Output: Features, Tracks, Depth

1: F ← FAST(L)

2: D ← ORB(F)

3: T ← LK-OpticalFlow(L_prev, L)

4: Z ← StereoDepth(L, R)

5: return (F, D, T, Z)

B. Factor-Graph Fusion Pseudocode

Algorithm 2: Multi-Sensor Fusion

Input: Visual factors, IMU, GPS, LiDAR, Odom

Output: Optimized state X

1: Initialize factor graph G

2: Add visual reprojection factors

3: Add IMU preintegration factors

4: Add GPS priors

5: Add LiDAR ICP factors

6: Add odometry factors

7: X ← iSAM2(G)

8: return X

C. Perception Stack Pseudocode

Algorithm 3: Perception Pipeline

Input: Image, Ego-pose

Output: Objects, Tracks

1: D ← CNN-Detector(Image)

2: For each detection d:

3:     t ← KalmanPredict()

4:     t ← KalmanUpdate(d, Ego-pose)

5: return Tracks

D. Dropout Recovery Pseudocode

Algorithm 4: Sensor Dropout Recovery

Input: Sensor status flags

Output: Fallback mode

1: if StereoLost:

2:     Use IMU + LiDAR

3: if LiDARSparse:

4:     Use StereoDepth

5: if GPSLost:

6:     Enable LoopClosure

7: return FallbackMode

APPENDIX C — System Parameters

Table VII — AUROADAS System Parameters

Component	**Parameter**	**Value**
Stereo	Baseline	12 cm
LiDAR	Range	100 m
IMU	Noise Density	0.002 rad/s/√Hz
GPS	Rate	5 Hz
Odometry	Resolution	2048 ticks/rev
CNN Detector	Confidence	0.45
ICP	Max Iterations	40

APPENDIX D — Transformer Mathematical Derivations

This appendix provides the mathematical foundations of the transformer-enhanced perception module used in AUROADAS. The derivations formalize Hilbert-guided sparse local attention, neighbor-aware token reduction, and semantic factor integration into the factor-graph backend.

A. Hilbert-Guided Sparse Local Attention

Let the convolutional stem produce a feature map

$$F \in \mathbb{R}^{H \times W \times C},$$

which is flattened into a sequence of tokens

$$T = \{t_1, t_2, \dots, t_N\},\ t_i \in \mathbb{R}^C.$$

Instead of raster ordering, AUROADAS applies **Hilbert curve ordering**, producing a locality-preserving permutation:

$$T_H = \{t_{\pi(1)}, t_{\pi(2)}, \dots, t_{\pi(N)}\},$$

where $\pi(\cdot)$maps 2D coordinates to Hilbert indices.

1) Sparse Neighborhood Definition

For each token t_i, define a sparse neighborhood:

$$N(t_i) = \{t_j \mid | \pi(j) - \pi(i) | \leq k\},$$

where $k$is the neighborhood radius (typically 8–32).

2) Sparse Attention Computation

Given query, key, and value projections:

$$q_i = W_q t_i,\ k_j = W_k t_j,\ v_j = W_v t_j,$$

attention weights are computed only over the sparse Hilbert neighborhood:

$$\alpha_{ij} = \frac{\exp\left(q_i^\top k_j / \sqrt{d}\right)}{\sum_{t_m \in N(t_i)} \exp\left(q_i^\top k_m / \sqrt{d}\right)}.$$

The sparse attention output is:

$$\text{Attn}(t_i) = \sum_{t_j \in N(t_i)} \alpha_{ij}\, v_j.$$

3) Complexity Reduction

Full attention:

$$O(N^2)$$

Hilbert-guided sparse attention:

$$O(Nk)$$

Since $k \ll N$, complexity becomes **linear**, enabling real-time transformer inference on embedded hardware.

B. Neighbor-Aware Token Reduction

Token reduction is formulated as a salience-weighted selection problem.

1) Salience Score

Each token receives a salience score:

$$s_i = \lambda_f \parallel t_i \parallel + \lambda_m M_i + \lambda_s S_i,$$

where:

- $\parallel t_i \parallel$: feature magnitude
- M_i: motion cue from ego-pose (optical flow or IMU-derived)
- S_i: semantic prior (e.g., proximity to dynamic agents or lane boundaries)
- $\lambda_f, \lambda_m, \lambda_s$: learned weights

2) Token Selection

Define a threshold τsuch that:

$$\mathcal{T}_{\text{keep}} = \{t_i \mid s_i \geq \tau\},$$

$$\mathcal{T}_{\text{merge}} = \{t_i \mid s_i < \tau\}.$$

Low-salience tokens are merged:

$$t_i' = \frac{1}{\mid N(t_i) \mid} \sum_{t_j \in N(t_i)} t_j.$$

This preserves semantic structure while reducing token count.

3) Complexity Impact

If the original token count is $N$and the reduced count is N':

$$N' = \rho N,\ 0 < \rho < 1,$$

then sparse attention complexity becomes:

$$O(N'k) = O(\rho N k).$$

Empirically, AUROADAS achieves:

$$\rho \approx 0.35\text{–}0.55,$$

cutting transformer compute by **45–65%**.

APPENDIX E — Transformer Module Pseudocode

This appendix presents high-level pseudocode for the AUROADAS transformer perception module, including Hilbert-guided sparse attention and neighbor-aware token reduction.

```
Algorithm 1: AUROADAS Transformer Perception Module

Input:
    F         // Conv stem feature map, shape H x W x C
    EgoPose   // Ego pose and motion cues
    SemPri    // Semantic priors (lanes, ROIs, etc.)

Output:
    SemEmb    // Semantic embeddings
    Det       // Object detections
    Seg       // Segmentation masks

// Tokenization along Hilbert curve
T <- HilbertTokenize(F)           // T = {t_1, ..., t_N}
// Compute salience scores
for each token t_i in T do
    mag_i <- ||t_i||              // Feature magnitude
    mot_i <- MotionCue(EgoPose, t_i)
    sem_i <- SemanticCue(SemPri, t_i)
    s_i   <- λ_f * mag_i + λ_m * mot_i + λ_s * sem_i
end for
// Neighbor-aware token reduction
T_keep   <- {}
T_merge  <- {}
for each token t_i in T do
    if s_i >= τ then
        T_keep <- T_keep ∪ {t_i}
    else
        T_merge <- T_merge ∪ {t_i}
    end if
end for
// Merge low-salience tokens into neighbors
for each token t_i in T_merge do
    N_i <- HilbertNeighborhood(T, i, k)
    t_i_merged <- (1 / |N_i|) * Σ_{t_j ∈ N_i} t_j
    AssignMergedToken(T_keep, t_i_merged)
end for
// Sparse local attention over Hilbert neighborhoods
for each token t_i in T_keep do
    q_i <- W_q * t_i
    N_i <- HilbertNeighborhood(T_keep, i, k)
    for each t_j in N_i do
        k_j <- W_k * t_j
        v_j <- W_v * t_j
        e_ij <- (q_i^T * k_j) / sqrt(d)
    end for
    α_ij <- Softmax_{j ∈ N_i}(e_ij)
    Attn_i <- Σ_{t_j ∈ N_i} α_ij * v_j
end for
// Transformer head for semantic outputs
SemEmb <- SemanticHead(Attn)
Det    <- DetectionHead(Attn, SemEmb)
Seg    <- SegmentationHead(Attn, SemEmb)
return SemEmb, Det, Seg
```

C. Semantic Factor Integration into the Factor Graph

Transformer outputs produce semantic cues:

$$\mathcal{S} = \{s_1, s_2, \ldots, s_M\},$$

including:

- segmentation boundaries
- dynamic-agent embeddings
- lane semantics
- obstacle surfaces

These cues are incorporated as **soft semantic factors**.

1) Semantic Boundary Factor

Let b_i be a boundary point predicted by the transformer, and p_t the ego-pose at time t.

Define a semantic residual:

$$r_b = \| f(p_t) - b_i \|,$$

where $f(p_t)$ projects the ego-pose into the semantic map.

The factor cost is:

$$\| r_b \|_{\Sigma_b}^2.$$

2) Dynamic-Agent Motion Factor

Let d_i be a dynamic agent embedding and $\hat{v}_i$ its predicted velocity.

Residual:

$$r_d = \| v_i - \hat{v}_i \|,$$

where v_i is the velocity estimated by the factor graph.

3) Semantic Lane Factor

Lane centerline L predicted by the transformer yields:

$$r_L = \mathrm{dist}(p_t, L),$$

penalizing lateral drift.

4) Combined Optimization

The full optimization becomes:

$$X^* = \arg\min_X \sum_{i \in \text{geom}} \| r_i \|_{\Sigma_i}^2 + \sum_{j \in \text{semantic}} \| r_j \|_{\Sigma_j}^2.$$

Semantic factors improve robustness under:

- occlusion
- motion blur
- LiDAR sparsity
- GPS dropout

D. End-to-End Latency Analysis

Let:

- T_{conv}: convolutional stem
- T_{attn}: sparse attention
- T_{reduce}: token reduction
- T_{fusion}: factor-graph update

Total latency:

$$T_{\text{total}} = T_{\text{conv}} + T_{\text{attn}} + T_{\text{reduce}} + T_{\text{fusion}}.$$

Empirical measurements on Jetson Orin AGX:

$$T_{\text{conv}} = 6\text{–}8 \text{ ms},$$

$$T_{\text{attn}} = 5\text{–}7 \text{ ms},$$

$$T_{\text{reduce}} = 2\text{–}3 \text{ ms},$$

$$T_{\text{fusion}} = 6\text{–}8 \text{ ms}.$$

Total:

$$T_{\text{total}} = 22\text{–}28 \text{ ms},$$

meeting real-time ADS requirements.

E. Summary

This appendix formalizes the mathematical foundations of the transformer-enhanced perception module in AUROADAS. Hilbert-guided sparse attention and neighbor-aware token reduction provide global reasoning with embedded-friendly complexity, while semantic factors strengthen the factor-graph backend under degraded sensing. These derivations demonstrate how efficient transformer architectures can be integrated into real-time autonomous driving systems.

APPENDIX F — Factor-Graph Equations

This appendix summarizes the core factor-graph formulation used in AUROADAS, including geometric, inertial, and semantic factors.

A. State Vector

The state at time $t$is:

$$X_t = \{R_t, p_t, v_t, b_t^a, b_t^g\},$$

where:

- R_t: rotation
- p_t: position
- v_t: velocity
- b_t^a: accelerometer bias
- b_t^g: gyroscope bias

B. General Factor Form

Each measurement factor is:

$$z_i = h_i(X) + \epsilon_i,$$

with residual:

$$r_i(X) = h_i(X) - z_i,$$

and cost:

$$\| r_i(X) \|_{\Sigma_i}^2.$$

The full optimization problem:

$$X^* = \arg\min_X \sum_{i \in \mathcal{G}} \| r_i(X) \|^2_{\Sigma_i} + \sum_{j \in \mathcal{S}} \| r_j(X) \|^2_{\Sigma_j},$$

where:

- $\mathcal{G}$: geometric/inertial factors
- $\mathcal{S}$: semantic factors

C. Geometric & Inertial Factors

1) IMU Preintegration Factor

Preintegrated IMU measurements:

$$\Delta R_{ij}, \Delta v_{ij}, \Delta p_{ij}$$

Residual:

$$r_{\text{IMU}} = \begin{bmatrix} \text{Log}\left(\Delta R_{ij}^{\top}(R_i^{\top} R_j)\right) \\ R_i^{\top}(v_j - v_i - g\Delta t_{ij}) - \Delta v_{ij} \\ R_i^{\top}\left(p_j - p_i - v_i \Delta t_{ij} - \frac{1}{2} g \Delta t_{ij}^2\right) - \Delta p_{ij} \end{bmatrix}.$$

2) LiDAR ICP Factor

Relative pose from ICP:

$$T_{ij}^{\text{LiDAR}} \in SE(3),$$

Residual:

$$r_{\text{LiDAR}} = \text{Log}\left(\left(T_i^{-1} T_j\right)\left(T_{ij}^{\text{LiDAR}}\right)^{-1}\right).$$

3) Stereo Reprojection Factor

Landmark $l_k$observed in image:

$$u_k = \pi\left(R_t^{\top}(l_k - p_t)\right),$$

Residual:

$$r_{\text{stereo}} = u_k^{\text{meas}} - u_k^{\text{pred}}.$$

4) GPS Factor

GPS position:

$$z_{\text{GPS}} = p_t + \epsilon_{\text{GPS}},$$

Residual:

$$r_{\text{GPS}} = p_t - z_{\text{GPS}}.$$

5) Odometry Factor

Odometry velocity/yaw:

$$r_{\text{odom}} = \begin{bmatrix} v_t - v_{\text{odom}} \\ \psi_t - \psi_{\text{odom}} \end{bmatrix}.$$

D. Semantic Factors (Transformer-Derived)

1) Semantic Boundary Factor

Boundary point $b_i$from segmentation:

$$r_b = \| f(p_t) - b_i \|,$$

where $f(p_t)$projects ego-pose into the semantic map.

2) Dynamic-Agent Motion Factor

Transformer-predicted velocity $\hat{v}_i$:

$$r_d = v_i - \hat{v}_i.$$

3) Lane Semantic Factor

Lane centerline L:

$$r_L = \text{dist}(p_t, L).$$

E. iSAM2 Incremental Update

The factor graph is solved incrementally using iSAM2:

1. **Linearization** around current estimate $X^{(k)}$
2. **Solve** for increment δXvia:

$$H\delta X = -g,$$

where:

- H: approximate Hessian
- g: gradient

3. **Update**:

$$X^{(k+1)} = X^{(k)} \oplus \delta X.$$

Semantic factors are treated as additional residuals contributing to H and g, improving robustness under degraded sensing.